



Technological relatedness, boundary-spanning combination of knowledge and the impact of innovation: Evidence of an inverted-U relationship

Jongtae Shin ^{*}, David Jalajas ¹

Long Island University, CW Post Campus, 720 Northern Blvd. Brookville, NY 11548, United States

ARTICLE INFO

Keywords:

Technological relatedness
Diversification
Knowledge flow
Knowledge combination

ABSTRACT

Our paper proposes that corporate technological relatedness, or the degree to which business units within a corporation utilize similar technological knowledge, has both positive and negative effects on corporate R&D activities. On the one hand, business units that employ similar technological knowledge have better absorptive capacity to source knowledge from each other. On the other hand, a higher level of technological relatedness means that each business unit possesses fewer opportunities to gain new knowledge not known to other units, thus promoting path dependence to each other. Using a patent data analysis of 201 firms in R&D-intensive industries, we examine the effects of corporate technological relatedness on within-firm knowledge flow, boundary-spanning combinations of prior knowledge, and innovation impacts.

© 2010 Elsevier Inc. All rights reserved.

1. Introduction

In strategic management theory, the resource-based view suggests that firms should avoid operating business portfolios that require vastly different kinds of resources and capabilities. According to this view, related diversifiers that exploit identical resources and capabilities across different businesses perform better than unrelated diversifiers unable to create synergistic effects. Thus, this view recommends that firms develop core competences to excel in a small number of aspects along value chain activities and join industries where they can exploit those competences. This theoretical argument has generated many empirical studies that vary with respect to the kinds of resources examined and the methodological approaches taken to measure relatedness among different business units.

Recent studies have suggested that achieving economies of scope based on technological knowledge is one of the most appropriate reasons for diversification (Miller, 2004; Robins & Wiersema, 1995). The costs associated with additional application of technological knowledge within a corporate boundary are trivial. At the same time, transferring knowledge through a market mechanism is cumbersome, since devising contracts sophisticated enough to prevent opportunistic behavior by both knowledge seller and buyer is difficult. In line with this argument, recent empirical studies have demonstrated that firms creating synergy based on technological knowledge exhibit superior performance than those do not (Miller, 2006; Silverman, 1999; Tanriverdi & Venkatraman, 2005). These studies imply that corporation with business units of similar technological expertise will perform better than those with business units that pursue different technological paths.

Although the similarity of technological expertise among a corporation's subunits may create synergy in R&D activities, excessive technological overlap may undermine a firm's R&D performance. Within the context of overseas R&D networks of multinational corporations, Song and Shin (2008) argued that R&D units have little incentive to source knowledge from each other when they have similar technological capabilities, thereby learning little from each other. On the other hand, innovations

^{*} Corresponding author. Tel.: +1 516 299 3175; fax: +1 516 299 3917.

E-mail addresses: jongtae.shin@liu.edu (J. Shin), djalajas@liu.edu (D. Jalajas).

¹ Tel.: +1 516 299 2292; fax: +1 516 299 3917.

generated by combining diverse technological expertise tend to have a wider impact on ensuing technological evolution, than do those originating from a narrow range of technological expertise (Rosenkopf & Nerkar, 2001; Yayavaram & Ahuja, 2008). These studies imply that corporations with both similar and diverse subunits, in terms of technological expertise, may exhibit higher R&D performance than do those with little technological variation among their subunits.

In this paper, we investigate how technological relatedness among corporate subunits, or the degree to which corporate subunits utilize similar technological knowledge, affect a firm's innovation activities and performance. We hypothesize a positive linear effect of technological relatedness among the subunits on knowledge flow within a corporate boundary due to superior absorptive capacity (Cohen & Levinthal, 1990) among the subunits. As technological relatedness facilitates knowledge flow among subunits, these subunits will have more opportunities to learn from each other. However, a high level of technological relatedness entails a negative effect on corporate R&D performance in that subunits are less likely to introduce new knowledge elements to each other. We argue that at an extremely high degree of technological relatedness, such negative effect prevails over the positive effect of increasing knowledge flow. We support this argument by empirically showing that technological relatedness has an inverted-U relationship with the frequencies of technological boundary-spanning combination of knowledge elements and with R&D performance.

Our paper uses patent data to measure within-firm knowledge flow and the impact of innovations created by firms. In addition, we employ Silverman's (1996) data that link patent classes and SIC industries to construct technological relatedness measures, and to assess boundary-spanning innovation. We employ a robust generalized linear model and negative binomial regression to investigate the effects of technological relatedness. Statistical findings based on 201 firms in R&D intensive industries support the argument that technological relatedness has both positive and negative effects on corporate R&D activities.

2. Theory and hypotheses

Theoretically, the resource-based view (Barney, 1991; Wernerfelt, 1984) has produced a compelling argument regarding how relatedness among different businesses may enhance a firm's performance. According to the resource-based view, firms sharing strategic resources among several businesses have advantages of scope economies (Teece, 1980). Economies of scope refer to the cost advantages arising when a firm utilizes a particular resource across several businesses (Panzar & Willig, 1981). By sharing significant resources across different business activities, firms with related diversification can increase revenues while saving potential costs associated with developing and applying new types of resources (Lubatkin & Chatterjee, 1994).

Although economies of scope are an important motive for diversification, it alone does not explain why firms resort to diversification instead of licensing or joint venture. Firms might be able to take advantage of economies of scope better by licensing assets to third parties or by sharing them with partners through joint ventures, when the bureaucratic costs associated with diversification are non-trivial. Thus, diversification becomes an attractive option when assets with potential economies of scope cannot be transferred to third parties by market mechanisms (Teece, 1980).

In general, firms face a high level of difficulty in transacting technological knowledge in the market (Teece, 1980; Von Hippel, 1994). When two independent parties buy and sell technological knowledge in the market, they should be concerned about the possibility of opportunistic behavior by each other. Without identifying the contents of technological knowledge, buyers would not be able to assess the potential value of a seller's technological knowledge. However, sellers would not reveal the contents of technological knowledge prior to transaction because doing so may incur opportunistic behavior by the buyers. Therefore, transaction costs associated with selling technological knowledge in the market may exceed the bureaucratic costs associated with sharing technological knowledge through diversification.

Technological relatedness, or the similarities among business units in terms of technological expertise, will become increasingly higher as corporations continue to diversify in order to exploit technological knowledge in house. Robins and Wiersema (1995) found that corporate performance is higher for firms that have diversified into technologically-related industries than for those that diversified into technologically-unrelated industries. Indeed, corporations with a higher degree of technological relatedness among subunits may have several advantages that contribute to a firm's R&D activities. Business units with similar technological expertise will have higher absorptive capacity to learn from each other (Cohen & Levinthal, 1990). According to Lane and Lubatkin (1998), the ability of an actor to learn from the other is high when two actors have similar knowledge base. Szulanski (1996) showed that a lack of absorptive capacity among business units is one of the major impediments to within-firm knowledge transfer. Darr and Kurtzberg (2000) showed that two stores transfer knowledge with each other more frequently when the strategic similarities of the stores are high. Song and Shin (2008) also argued that multinational corporations are more likely to source knowledge from their host countries when the firms' technological profiles are similar to those of the host countries. Thus, corporate subunits that have built similar technological expertise will have higher absorptive capacity to learn from each other, which can enhance knowledge flow within a corporation.

Technological relatedness will not only enhance absorptive capacity among business units, but will also increase the motivation of each subunit to search for technological solutions within corporate boundary. The sizeable costs associated with search beyond organizational boundaries often directs a firm's primary search efforts to the body of knowledge within the corporation (Rosenkopf & Nerkar, 2001; Stuart & Podolny, 1996). It is easier to transfer complex knowledge difficult to codify within the organizational boundary, than through the contexts of markets (Kogut & Zander, 1992). Transferring valuable knowledge across firms risks opportunistic behavior between knowledge senders and recipients (Grant, 1996; Teece, 1980). As a result, Singh (2005) found that within-firm knowledge flow is high because individual collaborative networks are more likely to be created within firms rather than across firms. Thus, a subunit is more likely to attempt to search for a solution in other subunits than outside the corporate boundary, if those subunits are perceived as having dealt with similar technological problems before and thus have a potential solution.

The above discussion suggests that technological relatedness among different businesses within a corporation will facilitate knowledge flow within a corporate boundary. Technological subunits with similar technological problems are likely to develop higher absorptive capacity in relation to each other. Given that these subunits will tend to perform local searches (Levinthal & March, 1993; March & Simon, 1958) both within organizational and technological boundaries, a higher absorptive capacity among subunits will increase the likelihood of knowledge flow among them. Thus,

H1. Technological relatedness among corporate subunits is positively associated with the degree of knowledge flow within corporation.

Technological relatedness among subunits is also likely to affect the degree to which they implement technological boundary-spanning combination of knowledge. Technological boundary-spanning combination of knowledge is the integration of knowledge elements from two or more technological domains that are distinct from each other (Rosenkopf & Nerkar, 2001; Yayavaram & Ahuja, 2008). Technological boundary spanning may occur as a result of a subunit's efforts to combine its technological knowledge with those of other subunits. Alternatively, the subunit may combine its knowledge elements with those of third parties outside the corporate boundary either directly or indirectly through other subunits that outsource knowledge from the third parties. Thus, technological boundary spanning can arise within or across organizational boundary (Rosenkopf & Nerkar, 2001).

A firm's ability to produce boundary-spanning innovation is one of the most important sources of sustainable competitive advantage, since it is deeply embedded in non-obvious routines of working relationships among individuals and business units (Henderson & Cockburn, 1994; Kogut & Zander, 1992). Henderson and Cockburn (1994) showed that in pharmaceutical industries, firms' research productivities are significantly explained by their capabilities to combine knowledge from distinct scientific disciplines such as molecular biology, physiology, or biochemistry. Breakthrough innovations that combine knowledge from multiple technological domains are often rewarded by abnormal returns from product markets (Fleming, 2002; Utterback, 1994).

Corporate subunits with a higher level of technological relatedness should be able to better source knowledge from each other as argued in Hypothesis H1. However, the very higher level of technological relatedness indicates less distinctiveness in terms of knowledge sets among subunits. Thus, in a corporation with a high level of technological relatedness, the probability of a subunit learning a novel kind of knowledge from other subunit will be low despite frequent knowledge exchanges among the subunits. Under an extremely high level of technological relatedness, knowledge flow among subunits will not be able to trigger boundary-spanning combinations of technological knowledge. Other things being equal, innovations produced under such condition would be characterized largely as marginal extensions of the subunits' preceding R&D efforts.

When corporate subunits have distinct knowledge sets, each subunit can increase the opportunity to combine knowledge elements from multiple technological domains by learning from other subunits. However, as the subunits' knowledge sets become more dissimilar, predicting the usefulness of each other's knowledge would become more problematic and causal ambiguity associated with knowledge transfer will become stronger (Tyre & Von Hippel, 1997; Winter, 1995). Thus, the difficulty of learning associated with an extremely low level of technological relatedness will prevent each unit from grasping the opportunities to produce boundary-spanning innovations.

Corporate subunits are more likely to combine knowledge elements that span technological boundaries when technological relatedness among them is at a moderate level. Learning is more effective when knowledge acquirers are familiar with the contexts of new knowledge to some degree (Phene, Fladmoe-Lindquist, & Marsh, 2006). Mowery, Oxley, & Silverman (1998) showed that firms are more likely to form an alliance when their technological capabilities were moderately overlapped. Collaboration among R&D units should be effective when they have knowledge in common and at the same time, a non-overlapping social network to tap into different knowledge bases (Reagans & Zuckerman, 2001). Similarly, Yayavaram and Ahuja (2008) suggested that a firm equipped with multiple clusters of knowledge elements with sparse ties among them has both the opportunity to search broadly and the ability to incorporate new elements into its knowledge base. When subunits have common knowledge elements, each subunit will be able to evaluate and learn technological knowledge from other subunits. At the same time, each subunit owns technological knowledge or skills that are distinct from those of other subunits, and each has the ability to absorb knowledge outside the corporate boundary that may not be comprehensible to other subunits. In such case, two subunits equipped with both technological overlap and divergence will be able to play a bridging role for each other by introducing and transferring new knowledge elements.

Technological boundary-spanning activities should be less problematic when they require collaboration among corporate subunits only, rather than entail cooperation outside the corporate boundaries. Von Hippel (1994) suggested that when innovation requires distinct knowledge sets, each of which is held by different actors, the R&D activities should be either subdivided into small tasks among the actors, or performed by one actor and then transferred to another in iterative mode. Such subdivision of R&D activities, or iterative mode of working, should be less complicated to carry out when actors can work out the problem of opportunism under the same corporate hierarchy (Williamson, 1975) and communicate efficiently based on shared common languages and cultural codes (Arrow, 1974; Kogut & Zander, 1992; Sorenson, Rivkin, & Fleming, 2006). Thus,

H2. Technological relatedness among corporate subunits has an inverted-U relationship with the degree of a subunit's technological boundary-spanning combination of knowledge.

George, Kotha and Zheng (2008) showed that firms equipped with depth and breadth in technological capabilities can produce innovations that are likely to have a high impact on future innovations. According to Katila and Ahuja (2002), a firm's R&D performance critically depends on balanced search activities in terms of depth and scope. Increasing search depth, or the extent to

which a firm reuses its existing technological knowledge, provides for more reliable and predictable problem solving and deepens the firm's established technological expertise (Ghemawat & Spence, 1985; Zahra & George, 2002). As firm repeatedly exploits the current technological expertise, however, it will eventually reach a threshold beyond which it will find difficulty detecting additional opportunities of innovation (Lei, Hitt & Bettis, 1996). Therefore, search scope, or the extent to which a firm explores diverse fields of technological knowledge, should complement search depth (Ahuja & Lampert, 2001). Just as extreme search depth incurs negative consequences, extremely wide scope makes it difficult for firms to integrate new knowledge into the established technological expertise in a reliable manner. As a result, search techniques balanced between depth and scope should be more effective in generating high-impact innovation than those emphasizing only one factor.

When corporate subunits have dealt with highly similar technological issues and built overlapping technological capabilities, knowledge flow from one subunit to another may increase technological depth at the cost of scope for the knowledge recipient. This is because incoming knowledge is likely to trigger reuses of the recipient's existing knowledge only, without introducing new knowledge elements (Ahuja & Lampert, 2001). On the other hand, when technological relatedness among subunits is extremely low, the subunits will operate as separate islands of distinct technological expertise, without knowledge flow among them, since they do not possess capacity to absorb knowledge from each other (Lane & Lubatkin, 1998). If so, a corporation with extremely low technological relatedness may seem to have the advantage of high level of technological scope, but subunits are unable to exploit the advantage of the wide scope. As a result, subunits with little technological relationship to each other will, in fact, exhibit disadvantages in terms of both search depth and scope. Thus, innovations resulting from both cases of exceedingly high or low level of technological relatedness are less likely to have an impact on subsequent innovations.

A moderate degree of technological relatedness within a corporation should offer a more advantageous combination of depth and scope to a subunits' R&D activities, than extremely high or low degree of corporate technological relatedness. Subunits bearing both a similarity and dissimilarity in terms of technological expertise will be able to increase knowledge flow among each other and, at the same time, add new knowledge elements to each other. A dynamic flow of knowledge that triggers both reuses of extant knowledge and learning of new knowledge elements should be able to increase the depth and scope of each subunit's R&D activities. Interactions between the depth and scope of subunits' R&D activities will then enhance R&D performance at corporate level (Katila & Ahuja, 2002; Gupta, Smith, & Shalley, 2006). Thus,

H3. Technological relatedness among corporate subunits has an inverted-U relationship with the degree of impact generated by corporate innovations.

3. Methodology

3.1. Sample

The empirical analysis reported here tests hypotheses based on a number of variables constructed from patent data. We use USPTO website in order to collect various information associated with patent filings such as application years, citations, and US patent classes. In addition, we use Silverman's (1996) data in order to calculate technological relatedness for each firm. Using Silverman's data requires the observation period to be close to 1993, the year on which Silverman's data was constructed. Locating observation period far from this period may cause less accurate reflection of patent–industry linkages.

The sample consists of 201 public firms in R&D-intensive industries from the DISCLOSURE database. In order to define an R&D-intensive industry, we use the approach adopted by Hadlock, Hecker and Gannon (1991). Hadlock, Hecker and Gannon (1991), using the occupational employment data released by Bureau of Labor Statistics, identified an industry as R&D intensive if the proportion of R&D workers is significantly higher than the average proportion of all industries. We eliminated firms with single SIC code and those with multiple SIC codes, if all the secondary codes correspond to industries that do not generate any patents, according to Silverman's data, or if there was any change in SIC code profiles between 1993 and 1998. Firms without patent filing activities at all between 1994 and 1998 were also eliminated from the final sample. Firms that are inherently not active in patent filing may bias the effects of independent variables.

3.2. Variables

The analysis goal in this paper is to examine the effects of technological relatedness among different businesses on the frequency of knowledge flow within corporation, the frequency of novel combinations among technological knowledge, and corporate R&D performance. We first calculated the Euclidean distance representing the dissimilarity between two industries i and j in terms of patent classes applicable to the industries based on Silverman's (1996) data. Based on the Canadian patent officers' industry category assignments on each patent and the concordance table on US SIC and Canadian SIC categories, Silverman (1996) derived the probability that any patent in patent class c will be utilized in industry i . We reorganize his data in order to obtain the profiles of patent classes for each industry and calculated Euclidean distance as follows.

$$\sqrt{\sum_{c=1}^n (P_{ic} - P_{jc})^2}$$

Where i, j = industries, c = patent class, n = the total number of patent classes assigned to industry i or j by Canadian patent officers.

Then, we identify SIC code profiles that sample firms had at the end of the year 1993 from the DISCLOSURE database. For each firm, we calculate Euclidean distance of every possible pair of SIC industry i and j as explained above, and obtain the average distance by dividing sum of Euclidean distances by the number of possible pairs. The average Euclidean distance measures technological relatedness in the opposite direction. In other words, a higher value of Euclidean distance indicates that the focal corporation has low degree of technological relatedness among different businesses.

In order to test Hypothesis H1, we measure the degree of knowledge flow within corporate boundary by proportion of self-citations made by firm's patents filed between 1994 and 1998. Each patent includes citations to earlier patents that make it possible to identify the origins and destinations of knowledge flow. Patent applications are often filed at subsidiary levels rather than at corporate level. Therefore, we identify subsidiaries of each sample firm from the DISCLOSURE database and patents assigned to corporate headquarter or to these subsidiaries and count the number of self-citations in which the corporate headquarter or its own subsidiaries made references among each other. Then we obtain the proportion of self-citations by dividing the number of self-citations with the total number of citations made by the firm between 1994 and 1998.

Testing Hypothesis H2 requires identification of patents resulting from novel combinations of multiple technological domains. One way to measure boundary-spanning innovation is to count the number of patents that contain citations to previous patents from a variety of patent classes. The pitfall in this approach is that in many cases, combinations of some different patent classes are widely practiced across firms rather than being unique to a particular firm. For instance, the technologies associated with sanitary equipments (International Patent Class: A47K) are often combined with the technologies associated with pipes and joints manufacturing (International Patent Class: F16L) to produce water supply equipments. Even though the two patent classes seem distinctive from each other in terms of International Patent Class (IPC) labels, it is inappropriate to consider such institutionalized combinations of technologies as being boundary spanning.

In order to measure the degree of boundary-spanning innovation, we identify combinations of prior knowledge that are relatively unlikely to occur at each patent level and calculate the proportion of such combinations for each firm. First, based on about 148,000 patents granted between 1990 and 1993 and 822 IPCs at subclass level identified in Silverman's (1996) data, we calculate Euclidean distance for every possible combination of two IPCs $c1$ and $c2$ ($882 \times 881 / 2 = 337,431$ pairs) as follows.

$$\text{Euclidean distance between } c1 \text{ and } c2 = \sqrt{\sum_{i=1}^n (P_{i,c1} - P_{i,c2})^2}$$

Where,

n = the total number of SIC industries to which patents in IPC $c1$ or $c2$ are assigned by Canadian patent examiners.

$P_{i,c1}$ = the number of patents in IPC $c1$ that were assigned to SIC industry i divided by the total number of patents in IPC $c1$.

Higher Euclidean distance indicates that knowledge elements in patent class $c1$ and those in patent class $c2$ are unlikely to be combined together for innovation. This calculation produces an "IPC-Euclidean distance" table in which every possible pair of IPCs is listed together with its Euclidean distance value.

Second, we identify every possible pair of IPCs of cited patents within each patent filed by sample firms between 1994 and 1998 and assigned Euclidean distance value to those pairs by looking up the IPC-Euclidean distance table. This identification processes based on 64,330 patents filed by sample firms produces the distribution of Euclidean distance values of 502,234 actual IPC pairs. The resulting distribution of Euclidean distance values ranges from 0.0247 to 1.3855 with first quartile ($Q1$) and third quartile ($Q3$) as 0.4591 and 0.7396 respectively. We classify an IPC pair as boundary-spanning combinations of prior knowledge if its Euclidean distance value is an outlier or higher than 1.1604 ($= Q3 + 1.5 * (Q3 - Q1)$). The total number of outliers is 1844 produced by 85 firms. We calculate the proportion of boundary-spanning combinations of prior knowledge for each firm by dividing the total number of novel combinations produced by the firm with the total number of combinations of citations as follows:

$$\frac{\sum_{p=1}^m NC_p}{\sum_{p=1}^m \frac{n_p(n_p-1)}{2}}$$

Where, n_p : the number of citations made by patent p , m : the number of patents filed by firm between 1994 and 1998. NC_p : the number of boundary-spanning combinations made by patent p .

In order to test Hypothesis H3, we operationalize R&D performance as the total number of citations received by firm's patents applied between 1994 and 1998. Patents often cited by other subsequent patents should represent impactful innovations that other firms recognize as effective solutions to the current technological problems and try to build their own technological capabilities upon. Patents issued earlier, however, are likely to be cited more than those issued later, simply because the former are exposed to the public longer. Therefore, we count the number of citations that each patent received for the duration of eight years subsequent to its application year.

We employ three distinct statistical analyses in order to test the hypotheses that require different dependent variables. In each analysis, we control for the total number of patents applied by each firm between 1994 and 1998 and the total number of patents issued to each firm until 1993. We also controlled for both firm size, measured as the logarithm of average total assets across three year periods, 1991–1993, and the number of SIC codes assigned to each firm.

Table 1

Summary of descriptive statistics.

	Mean	Std. dev.	2	3	4	5	6	7	8	9
1. Logarithm of total assets	12.50	2.37	0.25	0.26	0.71	0.43	0.21	0.10	0.21	0.35
2. Euclidean distance measure of technological relatedness	0.40	0.15	–	0.97	0.14	0.15	0.12	–0.17	–0.19	0.02
3. Euclidean distance measure of technological relatedness ²	0.18	0.16	–	–	0.13	0.16	0.14	–0.16	–0.19	0.01
4. Total number of SICs	3.92	1.84	–	–	–	0.29	0.19	0.03	0.18	0.23
5. Total number of patents filed between 1994 and 1998	317.81	1271.76	–	–	–	–	0.32	0.21	0.24	0.69
6. Total number of patents issued until 1993	691.21	2441.56	–	–	–	–	–	0.13	–0.01	0.64
7. Self-citation rate	0.09	0.08	–	–	–	–	–	–	0.15	0.21
8. Proportion of boundary-spanning combination of prior knowledge	0.07	0.11	–	–	–	–	–	–	–	0.21
9. Total number of citations received	8541.73	47,280.12	–	–	–	–	–	–	–	–

4. Results

Table 1 presents descriptive statistics of the variables. Hypothesis H1 proposed that technological relatedness is positively associated with within-firm knowledge flow. We employ robust generalized linear model (GLM) with binomial family and logit link in order to test the effect of technological relatedness on knowledge flow within corporation. The dependent variable is self-citation rate and the proportional dependent variable bounded between zero and one violates the assumption of normality in OLS regression. One may perform logit transformation on the proportional dependent variable and then employ OLS regression. However, this transformation results in missing values when observed values of the dependent variable include zero. Therefore, GLM analysis fits the purpose of current analysis better since the sample includes firms without self-citations (Papke & Wooldridge, 1996).

Table 2 summarizes the statistical findings from GLM test of Hypothesis H1. The base model is composed of control variables only. Eq. (1) in Table 2 adds the linear term of Euclidean distance measure of technological relatedness. The coefficient of the linear term in Eq. (1) is negative and significant (p -value < 0.05). These outcomes support Hypothesis H1, suggesting that the degree of technological relatedness among business units is positively associated with the degree of knowledge flow within corporation. Hypothesis H1 proposes that technological relatedness has linear effect on the dependent variable unlike Hypotheses H2 and H3. In order to make sure the linearity between technological relatedness and within-firm knowledge sourcing, we test the possibility of curvilinear effect in Eq. (2). When the quadratic term of Euclidean distance measure is added in Eq. (2), the linear term remains significant (p < 0.05) while the quadratic term is insignificant. Two control variables, total number of patents issued until 1993 (p < 0.05) and total number of patents filed between 1994 and 1998 (p < 0.01) are positive and significant as well.

Hypothesis H2 proposed that technological relatedness has an inverted-U relationship with technological boundary-spanning combination of knowledge. We use robust GLM model again in order to test the effect of technological relatedness on boundary-spanning combinations of prior knowledge, since the dependent variable is measured as proportion. The results of GLM analysis are presented in Table 3. As shown in Eq. (1) of Table 3, the linear term alone does not have significant effect on the proportion of boundary-spanning combinations. In Eq. (2) however, the coefficient of quadratic term is negative and significant (p < 0.05), while the coefficient of the linear term was positive and significant (p < 0.05), supporting Hypothesis H2. This result shows that corporations with moderate levels of technological relatedness will produce boundary-spanning combinations of prior knowledge more than do corporations with lower or higher levels of technological relatedness. The coefficient of total number of patents issued until 1993 is significant and negative in nonlinear model (p < 0.01). If the accumulated number of patents reflects firm's technological capabilities, then the negative coefficient may indicate that firms with high level of technological capabilities also have established technological trajectories (Stuart & Podolny, 1996), thus less motivated to experiment novel combinations of knowledge.

Hypothesis H3 proposed that technological relatedness has an inverted-U relationship with the degree of impacts generated by corporate innovations. We chose a negative binomial regression to test this hypothesis as the dependent variable is operationalized as the total number of citations received. A significant overdispersion problem indicates that Poisson regression is not appropriate in this analysis. The results in Table 4 suggest that the effect of technological relatedness on the total number of citations is similar to its effect

Table 2GLM analysis of self-citation rate ($n = 201$).

	Base model	Equation 1 (linear model)	Equation 2 (nonlinear model)
1. Intercept	0.2542 *** (0.0643)	0.2985 *** (0.0712)	0.2756 *** (0.0661)
2. Logarithm of total assets	0.0351 (0.0391)	0.0580 (0.0399)	0.0587 (0.0410)
3. Total number of patents filed between 1994 and 1998	0.00008 *** (0.00002)	0.00009 *** (0.00002)	0.00009 *** (0.00002)
4. Total number of patents issued until 1993	0.00003 * (0.000016)	0.00003 ** (0.000014)	0.00003 ** (0.000014)
5. The number of SICs	–0.0517 (0.0466)	–0.0553 (0.0441)	–0.0562 (0.0446)
6. Euclidean distance measure of technological relatedness		–1.5915 ** (0.6920)	–1.2664 ** (0.5235)
7. Euclidean distance measure of technological relatedness ²			–0.3257 (1.8165)

* Significant at p -value < 0.1.** Significant at p -value < 0.05.*** Significant at p -value < 0.01.

Table 3GLM analysis of boundary-spanning combination of prior knowledge ($n = 201$).

	Base model	Equation 1 (linear model)	Equation 2 (nonlinear model)
1. Intercept	0.1512 *** (0.0614)	0.1746 *** (0.0519)	0.1627 *** (0.0523)
2. Logarithm of total assets	0.0901 (0.0730)	0.1449 * (0.0793)	0.1967 *** (0.0743)
3. Total number of patents filed between 1994 and 1998	0.0001 *** (0.00004)	0.0001 *** (0.00004)	0.0001 ** (0.00004)
4. Total number of patents issued until 1993	−0.00007 (0.00004)	−0.00005 * (0.00003)	−0.00007 *** (0.00003)
5. The number of SICs	0.0571 (0.0801)	0.0523 (0.0758)	−0.0003 (0.0716)
6. Euclidean distance measure of technological relatedness		−3.9958 (2.9609)	4.2368 ** (2.0125)
7. Euclidean distance measure of technological relatedness ²			−4.1035 ** (1.94956)

* Significant at p -value < 0.1.** Significant at p -value < 0.05.*** Significant at p -value < 0.01.

on the boundary-spanning combination of prior knowledge. Although the coefficient of the linear term alone is not significant as shown in Eq. (1) of Table 4, the combined effects of both linear term ($p < 0.01$) and quadratic term ($p < 0.01$) clearly show that technological relatedness has inverted U relationship with the total number of citations received. In addition, the likelihood-ratio test suggests that the quadratic model has significantly higher explanatory power than the linear model ($p < 0.01$). This result indicates that innovations produced by corporations with moderate levels of technological relatedness are more influential than those produced by corporations with lower or higher levels of technological relatedness.

We calculate multipliers in order to estimate the effect of technological relatedness on the number of patent citations received by firms based on the nonlinear model estimated in Table 4. Fig. 1 shows the multiplier effects over the mean value of Euclidean distance measure of technological relatedness ($= 0.4$) with other variables held constant. In Fig. 1, the peak value of multiplier minus one of the inverted-U curve is 0.0159. This indicates that the firm at the optimal level of technological relatedness receives 1.6% more citations than those firms at average level of technological relatedness. Table 5 shows selected examples of firms with their Euclidean distance measures and the number of citations received per patent. For instance, the Euclidean distance measure for Altera Corporation with SIC 3674 (Semiconductors and Related Devices) and SIC 7372 (Prepackaged Software) is 0.5220 close to the value that gives the maximum number of citations according to Fig. 1.

5. Discussion and conclusion

This paper highlights the paradoxical effects of technological relatedness on innovation. Although technological relatedness may provide business units with a higher absorptive capacity to learn technological knowledge from each other, it reduces opportunities for business units to incorporate new knowledge elements from each other. The statistical findings of this paper support the hypotheses that technological relatedness has an inverted-U relationship with the frequencies of boundary spanning innovation and with the impacts of resulting innovations on subsequent technological evolutions.

Instead of adopting a simple dichotomy of unrelated vs. related diversification, this paper proposes that the two diversification approaches stand at opposite ends of a continuum and argue why undertaking diversification moves excessively in one direction may not be the best policy to pursue. Proper composition of a business portfolio not only provides the ability to exploit any single technological resource together, but also the opportunity to create impactful innovation by combining their distinct technological expertise.

The resource-based view suggests that a firm can enjoy economies of scope by expanding into new businesses in which it can exploit extant resources. Based on the theoretical advancement of knowledge-based view, this paper suggests that the right mixture of businesses within a corporation may not only increase the opportunity to exploit extant corporate resources but also enhances the likelihood of creating new technological resources. A firm's ability to produce innovations that span multiple

Table 4Negative binomial regressions of total number of citations received ($n = 201$).

	Base model	Equation 1 (linear model)	Equation 2 (nonlinear model)
1. Intercept	0.1315 *** (0.0519)	0.1014 ** (0.0497)	0.1131 ** (0.0505)
2. Logarithm of total assets	0.7439 *** (0.0685)	0.7365 *** (0.0681)	0.7539 *** (0.0655)
3. Total number of patents filed between 1994 and 1998	0.0008 *** (0.0001)	0.0008 *** (0.0001)	0.0008 *** (0.0002)
4. Total number of patents issued until 1993	−0.00003 (0.00004)	−0.00004 (0.00004)	−0.00003 (0.00004)
5. Self-citations rate	2.7106 ** (1.4177)	2.9997 ** (1.4591)	2.7656 * (1.4459)
6. The number of SICs	−0.5020 *** (0.0779)	−0.4951 *** (0.0771)	−0.5249 *** (0.0762)
7. Euclidean distance measure of technological relatedness		0.9518 (0.7825)	1.0675 *** (0.2624)
8. Euclidean distance measure of technological relatedness ²			−1.0171 *** (0.2509)
Log likelihood	−1529.0826	−1528.3209	−1522.6675

* Significant at p -value < 0.1.** Significant at p -value < 0.05.*** Significant at p -value < 0.01.

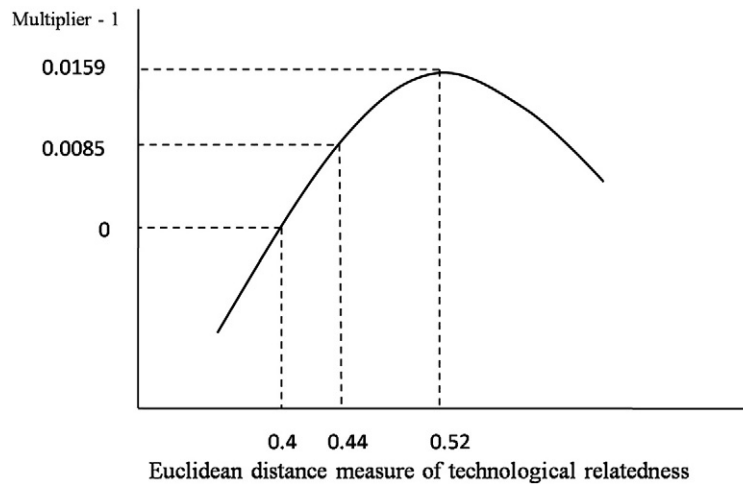


Fig. 1. Multiplier of the number of patent citations received.

technological domains is one of the most critical sources of sustained competitive advantage in R&D intensive industries (Henderson & Cockburn, 1994; Rosenkopf & Nerkar, 2001; Kogut & Zander, 1992). Pursuing extremely high or low levels of technological relatedness in business portfolios may prevent corporations from developing such boundary-spanning technological capabilities. The results of this paper imply that a portfolio of businesses that are moderately related to each other in terms of technological resources can provide more opportunities for each business unit to learn and innovate than those with extremely high or low level of technological relatedness.

Methodologically, this paper attempts to measure the relatedness of underlying resources among business units, instead of relying on hierarchical structure of SIC codes. Traditional measures of relatedness such as Herfindahl index (Montgomery, 1982) or Entropy index (Jacquemin & Berry, 1979) are constructed based on the assumption that two businesses that belong to the same SIC categories are more related to each other than otherwise. Such assumption is doubtful in that businesses sharing the same digit number at two or three digit level are not necessarily related to each other more than those businesses with different SICs. For example, the relatedness measures give a doubtful implication that SIC 3674 (Semiconductors and Related Devices) have more in common with SIC 3633 (Household Laundry Equipment) than with SIC 7372 (Computer Integrated System Design). In addition, these measures do not address the issue of what types of resources play a bridging role among business units. Based on Silverman (1996)'s data on the linkage between SIC industries and patent classes, this paper attempts to measure resource relatedness based on firm's underlying technological resources instead of using simple SIC hierarchies that may be loosely connected with corporate R&D activities.

The empirical findings of technological relatedness have an important managerial implication. While encouraging the development of central technological capabilities within corporation, top managers may also execute diversification policies that allow business units to have both commonalities and variations in terms of technological resources and capabilities. By doing so, firms can expect short-term returns based on economies of scope and aim at becoming a leader in technological evolutions in the

Table 5

The examples of corporations with technological relatedness and R&D performance.

Company	SIC codes	Euclidean distance measure	The number of citations received per patent
Watkins Johnson Co.	3812: Search & Navigation Equipment	0.1860	9
Moog Inc.	3559: Special Industry Machinery		
	3728: Aircraft Parts & Equipment	0.1907	11
	3724: Aircraft Engines & Engine Parts		
Pictoretel Corp.	3663: Radio & TV Communication Equipment	0.2457	6
	3669: Communication Equipment		
Cerprobe Corp.	3825: Electricity Meas Instruments	0.2682	10
	3674: Semiconductors & Related Devices		
Stanford Telecomm. Inc.	3663: Radio & TV Communication Equipment	0.2722	8
	3812: Search & Navigation Equipment		
Mentor Corp.	3841: Surgical & Medical Instruments	0.3409	12
	3842: Surgical Appliances and Supplies		
	3845: Electromedical Equipments		
Altera Corp.	3674: Semiconductors & Related Devices	0.5220	19
	7372: Prepackaged Software		
Intervice Inc.	3661: Telephone & Telegraph Apparatus	0.8515	11
	7373: Computer Integrated System Design		

long term. Thus, the results of this paper draw managers' attention to the possibility that firms can gain sustained competitive advantage from business portfolios open to both exploitation and exploration. However, such business portfolios will be of no use without managerial efforts to shape corporate culture and formal structures that encourage cooperation among R&D subunits (Ranft & Lord, 2002).

Effective R&D cooperation among business units and a fine balance between exploitation and exploration are undoubtedly dependent on formal organizational structure, corporate culture and managerial incentive systems (Ambos & Schlegelmilch, 2008; Hitt, Hoskisson, Johnson, & Moesel, 1996; Stieglitz & Heine, 2007; Teece, 2007). For instance, Argyres and Silverman's (2004) study on the relationship between organizational structure and innovation suggests that an optimal level of corporate technological relatedness should be accompanied by centralized R&D function. Tsai and Ghoshal (1998) stressed the critical role of social capital within firm as enhancing the likelihood of resource exchange among business units. Despite such importance, this paper did not incorporate internal control mechanisms or horizontal communication links as either primary independent variables or control variables due the availability of data. Future studies should investigate how diverse modes of R&D control mechanism interact with technological relatedness to influence R&D performance.

Future studies should also investigate how corporate technological relatedness influences product innovation. The number of citations received by firm's patents is one way to measure the impacts of firm's innovation but not the single most effective way to do so. Patent innovation is intermediate process to create new products. However, not every patent innovation leads to successful introduction of new products and firms often invent products without patenting in order to keep new secret recipes for innovation (Arundel & Kabla, 1998). As such, future empirical studies should investigate the validity of the results in this paper by examining product innovation as well.

References

- Ahuja, G., & Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22, 521–543.
- Ambos, B., & Schlegelmilch, B. B. (2008). Innovation and control in the multinational firm: A comparison of political and contingency approaches. *Strategic Management Journal*, 28(5), 473–486.
- Argyres, N. S., & Silverman, B. S. (2004). R&D, organizational structure and the development of corporate technological knowledge. *Strategic Management Journal*, 25(8/9), 929–958.
- Arrow, K. (1974). *The limits of organization*. New York: Norton.
- Arundel, A., & Kabla, I. (1998). What percentage of innovations are patented? Empirical estimates for European firms. *Research Policy*, 27, 127–141.
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.
- Cohen, W., & Levinthal, D. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35, 128–152.
- Darr, E. D., & Kurtzberg, T. R. (2000). An investigation of partner similarity dimensions on knowledge transfer. *Organizational Behavior and Human Decision Processes*, 82, 28–44.
- Fleming, L. (2002). Finding the organizational sources of technological breakthroughs: The story of Hewlett Packard's thermal ink jet. *Industrial and Corporate Change*, 11(5), 1059–1084.
- George, G., Kotha, R., & Zheng, Y. (2008). Entry into insular domains: A longitudinal study of knowledge structuration and innovation in biotechnology firms. *Journal of Management Studies*, 45(8), 1448–1474.
- Ghemawat, P., & Spence, A. M. (1985). Learning-curve spillovers and market performance. *Quarterly Journal of Economics*, 100, 839–852.
- Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17, 109–122.
- Gupta, A. K., Smith, K. G., & Shalley, C. E. (2006). The interplay between exploration and exploitation. *Academy of Management Journal*, 49, 693–706.
- Hadlock, P., Hecker, D., & Gannon, J. (1991, July). High technology employment: Another view. *Monthly Labor Review*, 26–30.
- Henderson, R. M., & Cockburn, I. M. (1994). Measuring competence? Exploring firm effects in pharmaceutical research. *Strategic Management Journal*, 15, 63–84.
- Hitt, M. A., Hoskisson, R. E., Johnson, R. A., & Moesel, D. D. (1996). The market for corporate control and firm innovation. *Academy of Management Journal*, 39(5), 1084–1119.
- Jacquemin, A. P., & Berry, C. H. (1979). Entropy measure of diversification and corporate growth. *Journal of Industrial Economics*, 27, 359–369.
- Katila, R., & Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6), 1183–1194.
- Kogut, B., & Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3, 383–397.
- Lane, J. P., & Lubatkin, M. (1998). Relative absorptive capacity and interorganizational learning. *Strategic Management Journal*, 19, 461–477.
- Lei, D., Hitt, M. A., & Bettis, R. (1996). Dynamic core competencies through meta-learning and strategic context. *Journal of Management*, 24(2), 248–270.
- Levinthal, D. A., & March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14, 95–112.
- Lubatkin, M., & Chatterjee, S. (1994). Extending modern portfolio theory into the domain of corporate diversification: Does it apply? *Academy of Management Journal*, 37(1), 109–136.
- March, J. G., & Simon, H. A. (1958). *Organizations*. New York: Wiley.
- Miller, D. J. (2004). Firms' technological resources and the performance effects of diversification: A longitudinal study. *Strategic Management Journal*, 25(11), 1097–1119.
- Miller, D. J. (2006). Technological diversity, related diversification, and firm performance. *Strategic Management Journal*, 27, 610–619.
- Montgomery, C. A. (1982). The measurement of firm diversification: Some new empirical evidence. *Academy of Management Journal*, 25, 299–307.
- Mowery, D., Oxley, J., & Silverman, B. (1998). Technological overlap and interfirm coordination: Implications for the resource-based view of the firm. *Research Policy*, 27, 507–523.
- Panzar, J., & Willig, R. (1981). Economies of scope. *The American Economic Review*, 71, 268–272.
- Papke, L. E., & Wooldridge, J. (1996). Econometric methods for fractional response variables with an application to 401(k) plan participation rates. *Journal of Applied Econometrics*, 11, 619–632.
- Phene, A., Fladmoe-Lindquist, K., & Marsh, L. (2006). Breakthrough innovations in the US biotechnology industry: The effects of technological space and geographic origin. *Strategic Management Journal*, 27, 369–388.
- Ranft, A. L., & Lord, M. D. (2002). Acquiring new technologies and capabilities: A grounded model of acquisition implementation. *Organization Science*, 13(4), 420–442.
- Reagans, R., & Zuckerman, E. W. (2001). Networks, diversity and productivity: the social capital of corporate R&D teams. *Organization Science*, 12(4), 502–517.
- Robins, J., & Wiersema, M. (1995). A resource-based approach to the multi-business firm: Empirical analysis of portfolio inter-relationships and corporate financial performance. *Strategic Management Journal*, 16(4), 277–299.
- Rosenkopf, L., & Nerkar, A. (2001). Beyond local search: Boundary-spanning exploration and impact in the optical disc industry. *Strategic Management Journal*, 22(4), 287–306.

- Silverman, S. S. (1996). *Technical assets and the logic of corporate diversification*. Ph.D: Dissertation, Hass School of Business, University of California, Berkeley, CA.
- Silverman, S. S. (1999). Technological resources and the direction of corporate diversification: toward an integration of the resource-based view and transaction cost economics. *Management Science*, 45(8), 1109–1124.
- Singh, J. (2005). Collaborative networks as determinants of knowledge diffusion patterns. *Management Science*, 51(5), 756–770.
- Song, J. Y., & Shin, J. T. (2008). The paradox of technological capabilities: A study of knowledge sourcing from host countries of overseas R&D operations. *Journal of International Business Studies*, 39(2), 291–303.
- Sorenson, O., Rivkin, J. W., & Fleming, L. (2006). Complexity, networks and knowledge flow. *Research Policy*, 36, 994–1017.
- Stieglitz, N., & Heine, K. (2007). Innovations and the role of complementarities in a strategic theory of the firm. *Strategic Management Journal*, 28(1), 1–15.
- Stuart, T. E., & Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17, 21–38.
- Szulanski, G. (1996). Exploring internal sickness: Impediments to the transfer of best practice within the firm. *Strategic Management Journal*, 17, 27–43.
- Tanriverdi, H., & Venkatraman, N. (2005). Knowledge relatedness and the performance of multibusiness firms. *Strategic Management Journal*, 26, 97–119.
- Teece, D. J. (1980). Economies of scope and the scope of the enterprise. *Journal of Economic Behavior and Organization*, 1, 223–245.
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350.
- Tsai, W., & Ghoshal, S. (1998). Social capital and value creation: The role of interfirm networks. *Academy of Management Journal*, 41(4), 464–476.
- Tyre, M. J., & von Hippel, E. (1997). The situated nature of adaptive learning in organizations. *Organization Science*, 8(1), 71–83.
- Utterback, J. M. (1994). Radical innovation and corporate regeneration. *Research-Technology Management*, 37(4), 10–21.
- Von Hippel E (1994). 'Sticky information' and the locus of problem solving: Implications for innovation. *Management Science*, 40, 429–439.
- Wernerfelt, B. (1984). A resource-based view of the firm. *Strategic Management Journal*, 5(2), 171–180.
- Williamson, O. (1975). *Markets and hierarchies: Analysis and antitrust implications*. New York: Free Press.
- Winter, S. G. (1995). Four Rs of profitability: Rents, resources, routines and replication. In Cynthia A. Montgomery (Ed.), *Resource-based and evolutionary theories of the firm: Towards a synthesis* (pp. 147–178). Dordrecht: Kluwer Academic Publishers.
- Yayavaram, S., & Ahuja, G. (2008). Decomposability in knowledge structures and its impact on the usefulness of inventions and knowledge-base malleability. *Administrative Science Quarterly*, 53, 333–362.
- Zahra, S. A., & George, G. (2002). Absorptive capacity: A review, reconceptualization, and extension. *Academy of Management Review*, 27, 185–203.